

HUJI Applicative Research Report — August 2026

6 high-potential paper(s) · Monthly · August 2026 · Generated August 07, 2026

Hebrew University researchers are pioneering breakthroughs across critical sectors, from next-generation genomics tools and AI-driven drug discovery platforms to novel materials and targeted therapeutics. This month's digest highlights innovations in precision diagnostics and drug development, particularly addressing chronic pain and cancer with high-potency compounds and AI-engineered immune receptors. Furthermore, advancements in smart materials for thermal management and sophisticated single-cell data analysis tools demonstrate significant commercial potential for investors. These technologies promise to address critical market gaps and accelerate progress in several high-value industries.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Avi Priel** — Drug Discovery, Neuroscience, Clinical
- **Roy Maman** — Materials, Imaging
- **Dina Schneidman-Duhovny** — Drug Discovery, Immunology, Software/AI, Synthetic Biology
- **Itamar Willner** — Synthetic Biology, Drug Discovery, Materials, Diagnostics
- **Mor Nitzan** — Drug Discovery, Software/AI, Genomics, Immunology

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method **35/50** for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells.

"New single-cell RNA sequencing offers cheaper, comprehensive data for precision diagnostics and drug discovery."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

ABOUT THIS RESEARCH

Researchers have developed a new single-cell sequencing technology called sc-rDSeq that captures a much wider range of RNA types, including non-polyadenylated RNAs, at a lower cost and higher density than current standards. By applying this to lung cancer, they demonstrated the ability to identify hidden cell subpopulations and drug-resistance mechanisms that traditional methods miss.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty

8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential

8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size

8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness

4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength

7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

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2. Carbamate-based cannabinoids uncover a conserved lipid-sensing pocket in nociceptive TRPA1 and TRPV1 channels. 33/50

"Potent carbamate cannabinoids open new avenues for non-opioid chronic pain treatments."

Main researcher: Avi Priel · avi.priel@mail.huji.ac.il.

The Institute for Drug Research (IDR), Department of Pharmacology, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Israel. Electronic address: avi.priel@mail.huji.ac.il.

Published in Pharmacological research · 2026-06

ABOUT THIS RESEARCH

Researchers have engineered new cannabinoid-based molecules that more effectively target TRPA1 and TRPV1 ion channels by adding a specific carbamate functional group. These modified compounds show significantly higher potency in activating these pain-sensing channels compared to natural CBD or CBG.

COMMERCIAL ANGLE

The development of these high-potency carbamate derivatives provides a structural roadmap for designing a new class of non-opioid analgesics for chronic pain management.

COMMERCIALISATION METRICS

Novelty

7/10

The research demonstrates significant chemical innovation by using rational design to transform known scaffolds into potent dual agonists, revealing a previously unknown conserved lipid-sensing mechanism. While it offers high target-validation value and potential for new drug classes, it builds incrementally on established phytocannabinoid chemistry rather than introducing an entirely new paradigm.

Commercial Potential

7/10

The research demonstrates successful rational design of small-molecule derivatives with enhanced potency and selectivity, representing a clear path toward a novel non-opioid analgesic lead series. However, the transition from structural discovery to a proprietary drug candidate requires significant validation of pharmacokinetic profiles and safety to reach high-tier commercial readiness.

Market Size

9/10

The research targets TRPA1 and TRPV1, two of the most high-value pathways in the multi-billion dollar non-opioid analgesic market. By developing potent dual-agonists, this work addresses a massive unmet medical need for chronic and acute pain management with significant commercial scalability.

Tech Readiness

3/10

The research is currently at the basic discovery/proof-of-concept stage, utilizing in vitro heterologous expression systems to validate molecular binding and potency. It lacks in vivo animal models, pharmacokinetic data, or toxicity profiling required to advance toward preclinical development.

IP Strength

7/10

The research demonstrates successful rational ligand design to create novel, highly potent carbamate-functionalized derivatives with improved selectivity, providing a clear pathway for composition-of-matter patent claims. While the chemical scaffolds are derivative of known phytocannabinoids, the specific structural modifications and the discovery of a conserved binding pocket offer significant defensibility for new non-opioid analgesic candidates.

[View in Dashboard Open Paper](#)

3. Metasurface platform for simultaneous and uncorrelated emissivity control over MWIR and LWIR spectral bands.

32/50

"Advanced metasurfaces enable independent control of thermal signatures for stealth and thermal management."

Main researcher: Roy Maman · Roy.maman@mail.huji.ac.il.

Institute of Applied Physics, The Faculty of Science, The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem, 91904, Israel.

Published in Light, science & applications · 2026 Jul 23

ABOUT THIS RESEARCH

Researchers have developed a metasurface capable of independently controlling how an object emits heat (thermal radiation) across two different infrared spectrums. This allows for the precise manipulation of thermal signatures and heat dissipation without one wavelength affecting the other.

COMMERCIAL ANGLE

High-value applications in stealth technology, advanced thermal management for electronics, and specialized non-invasive thermal imaging sensors.

COMMERCIALISATION METRICS

Novelty

7/10

The ability to achieve uncorrelated, simultaneous control over two distinct infrared bands (MWIR and LWIR) represents a significant functional leap over single-band or correlated metasurfaces, though it builds on existing plasmonic/dielectric resonator principles.

Commercial Potential

7/10

The ability to independently control MWIR and LWIR emissivity creates a high-value platform for thermal camouflage, energy-efficient coatings, and advanced sensing. While the technical application is clear, the score is tempered by the lack of provided data on scalability and manufacturing costs.

Market Size

7/10

The ability to independently control MWIR and LWIR emissivity is a platform-enabling capability with high value in defense (stealth/cloaking) and industrial thermal management, though it targets specialized high-end niches rather than a mass consumer market.

Tech Readiness

3/10

The paper describes the development of a metasurface platform for spectral control, which is characteristic of fundamental materials science and lab-stage proof-of-concept work. There is no evidence of integration into a commercial device, field testing, or scalable manufacturing processes.

IP Strength

8/10

The development of a platform for uncorrelated spectral control in two distinct bands represents a high-value 'platform-enabling' innovation with strong potential for broad patent claims across defense and industrial sensing.

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4. Progress in structure prediction and design of adaptive immune receptors. 30/50

"AI advances empower precision design of synthetic immune receptors for personalized cancer immunotherapy."

Main researcher: Dina Schneidman-Duhovny · dina.schneidman@mail.huji.ac.il.

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Published in Current opinion in structural biology · 2026 Jul 10

ABOUT THIS RESEARCH

The paper reviews advancements in using computational tools to predict the shapes and design the functions of immune receptors. This allows scientists to better understand how the body recognizes pathogens and how to engineer custom receptors for therapy.

COMMERCIAL ANGLE

Developing AI-driven platforms for the precision design of synthetic antibodies and T-cell receptors for personalized cancer immunotherapies.

COMMERCIALISATION METRICS

Novelty

4/10

Based on the title, this is a review paper summarizing 'progress' rather than a primary research article presenting a novel method or discovery. It synthesizes existing advancements in the field rather than introducing groundbreaking innovation.

Commercial Potential

8/10

Structure prediction for adaptive immune receptors is a platform-enabling capability that directly accelerates the design of high-affinity therapeutic antibodies and TCR-T cell therapies, representing a massive commercial market in precision medicine.

Market Size

9/10

The ability to predict and design adaptive immune receptors is a platform-enabling technology with massive addressable demand across biologics, personalized cancer vaccines, and autoimmune therapies.

Tech Readiness

3/10

The title indicates a review or research progress report on structure prediction and design, which is fundamentally a computational/foundational stage of development. It focuses on modeling and design principles rather than a validated clinical candidate or a commercialized diagnostic platform.

IP Strength

6/10

The work focuses on structure prediction and design for adaptive immune receptors, which provides a strong foundation for platform-based IP in biologics; however, without a specific novel algorithm or proprietary dataset mentioned, it likely represents incremental progress in an open-science field.

[View in Dashboard](#) [Open Paper](#)

5. Dynamic Switchable and Transient DNA Condensates Driven by Aptamer-Ligand or Ion-Nucleobase Bridged Complexes.

28/50

"Programmable DNA coacervates enable smart drug delivery and responsive biomaterials."

Main researcher: Itamar Willner

The Institute of Chemistry, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Small (Weinheim an der Bergstrasse, Germany) · 2026-06

ABOUT THIS RESEARCH

Researchers have developed a way to create DNA-based liquid droplets that can be programmed to form, dissolve, or change their properties on command using specific chemical triggers. These programmable 'coacervates' act as smart materials that can respond dynamically to their environment through molecular switching.

COMMERCIAL ANGLE

This technology enables the development of highly responsive, programmable drug delivery vehicles and smart biomaterials that release payloads only in response to specific biological or chemical signals.

COMMERCIALISATION METRICS

Novelty

8/10

The research introduces a highly innovative approach to controlling DNA phase separation through programmable, stimuli-responsive molecular bridges (aptamer-ligand and metal-ion/mismatch), moving beyond static coacervates toward dynamic, switchable materials. This represents a platform-enabling advancement for responsive drug delivery and programmable soft matter.

Commercial Potential

5/10

The technology demonstrates sophisticated fundamental control over DNA matter, but the abstract lacks specific application towards a high-value end-market such as targeted drug delivery or programmable diagnostics. While it serves as a platform-enabling material, the path to a specific, licensable product remains academically abstract rather than industrially defined.

Market Size

5/10

While the platform offers high scientific novelty for programmable materials, the specific application remains a fundamental research tool rather than a clearly defined high-value commercial market. The market size is currently constrained by its niche status in synthetic biology and materials science without a clear immediate path to large-scale therapeutic or industrial adoption.

Tech Readiness

3/10

The work describes fundamental proof-of-concept molecular design and basic biochemical behavior of DNA-based phase separation. It is currently at the basic research/early laboratory stage without evidence of integrated system testing or biological application scaling.

IP Strength

7/10

The use of programmable DNA architecture and specific chemical bridging (metal-ions/mismatched bases) provides a strong foundation for composition-of-matter patents. However, the inherent 'openness' of DNA nanotechnology and the high likelihood of prior art in DNA condensation and aptamer chemistry may complicate broad claim construction.

[View in Dashboard Open Paper](#)

6. Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck. 28/50

"AI framework streamlines single-cell data analysis, uncovering key biological signals for drug discovery."

Main researcher: Mor Nitzan · mor.nitzan@mail.huji.ac.il.

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Published in Cell systems · 2026-05

ABOUT THIS RESEARCH

This paper introduces bioIB, an AI framework that uses the information bottleneck method to distill complex single-cell RNA sequencing data into highly specific, interpretable gene clusters. By filtering out noise and focusing on relevant biological signals like disease states, it maps out the hierarchical relationships between different cell types and biological processes.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

COMMERCIALISATION METRICS

Novelty

6/10

The paper applies an established information theory framework (Information Bottleneck) to a well-trodden domain (scRNA-seq), representing an incremental methodological improvement rather than a fundamental scientific breakthrough. While the application to gene clustering and hierarchical structure is useful, it follows existing trends in deep learning-based dimensionality reduction and feature selection.

Commercial Potential

6/10

The framework presents a strong platform-enabling computational tool for drug discovery and biomarker identification, but its commercial value is currently tied to software/algorithm licensing rather than a standalone proprietary product. Its utility depends on integration into existing bioinformatics pipelines or single-cell analysis platforms.

Market Size

8/10

The framework acts as a platform-enabling technology for the massive and growing single-cell multi-omics market, offering high value for drug discovery and precision medicine. By optimizing signal extraction in complex datasets, it addresses a critical bottleneck in the multi-billion dollar bioinformatics and clinical diagnostics sectors.

Tech Readiness

4/10

The paper presents a computational framework and algorithmic methodology (bioIB) rather than a physical product or clinical tool. While it demonstrates utility on existing datasets, it remains at the fundamental software/algorithm development stage (TRL 3-4) rather than a validated industrial platform.

IP Strength

4/10

The innovation is primarily a computational framework based on the Information Bottleneck method, which is an established theoretical principle in machine learning, making the core algorithm difficult to patent. While the application to scRNA-seq is valuable, the defensibility is low as it likely falls under 'mathematical methods' and is susceptible to rapid replication by open-source community developments.

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